24. To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables.

Project

```
sanjana3027@DESKTOP-QJF5AGU:      ~$ mkdir Skill_Project
sanjana3027@DESKTOP-QJF5AGU:      ~$ cd Skill_Project
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project$ mkdir src include tests docs
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project$ cd src
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project/src$ cd ..
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project$ touch src/main.c
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project$ touch include/config.h
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project$ touch tests/test.sh
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project$ touch docs/architecture.txt
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project$ touch README.md
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project$ touch Makefile
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree  # version 2.1.3+pkg-5852, or
sudo apt  install tree  # version 2.3.1-1
See 'snap info tree' for additional versions.
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project$ sudo apt install tree
[sudo: authenticate] Password:
Installing:
  tree

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 172
  Download size: 53.5 kB
  Space needed: 125 kB / 1023 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 1s (39.6 kB/s)
Selecting previously unselected package tree.
(Reading database ... 43086 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
sanjana3027@DESKTOP-QJF5AGU:      ~/Skill_Project$ tree
.
├── Makefile
├── README.md
├── docs
│   └── architecture.txt
├── include
```

```
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ tree
.
├── Makefile
├── README.md
├── docs
│   └── architecture.txt
├── include
│   └── config.h
├── src
│   └── main.c
└── tests
    └── test.sh

5 directories, 6 files
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ nano src/main.c
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ nano include/config.h
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ nano Makefile
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ nano tests/test.sh
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ chmod +x tests/test.sh
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ nano README.md
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ make
gcc -Wall -Wextra -Iinclude src/main.c -o skill24
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ make run
./skill24
Skill 24 Repository Demo
Version: 1.0.0
Repository validation successful.
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ make test
./tests/test.sh
Running Skill 24 tests...
-------------------------
PASS: Repository validation test
PASS: Version test
-------------------------
All tests passed.
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ make clean
rm -f skill24
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ nano docs/architecture.txt
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ make
gcc -Wall -Wextra -Iinclude src/main.c -o skill24
sanjana3027@DESKTOP-QJF5AGU:~/Skill_Project$ ls
```

```
sanjana3027@DESKTOP-QJF5AGU:          :~/Skill_Project$ ls
Makefile  README.md  docs  include  skill24  src  tests
sanjana3027@DESKTOP-QJF5AGU:          :~/Skill_Project$ make run
./skill24
Skill 24 Repository Demo
Version: 1.0.0
Repository validation successful.
sanjana3027@DESKTOP-QJF5AGU:          :~/Skill_Project$ make test
./tests/test.sh
Running Skill 24 tests...
-------------------------
PASS: Repository validation test
PASS: Version test
-------------------------
All tests passed.
```

```
  GNU nano 8.7.1                                                src/main.c *
#include <stdio.h>
#include "../include/config.h"

int main() {

    printf("%s\n", PROJECT_NAME);
    printf("Version: %s\n", PROJECT_VERSION);
    printf("Repository validation successful.\n");

    return 0;
}
```

```
  GNU nano 8.7.1                                          include/config.h *
#ifndef CONFIG_H
#define CONFIG_H

#define PROJECT_NAME "Skill 24 Repository Demo"
#define PROJECT_VERSION "1.0.0"

#endif
```

```
  GNU nano 8.7.1                                                Makefile *
CC = gcc
CFLAGS = -Wall -Wextra -Iinclude

TARGET = skill24

SRC = src/main.c

all: $(TARGET)

$(TARGET): $(SRC)
        $(CC) $(CFLAGS) $(SRC) -o $(TARGET)

run: $(TARGET)
        ./$(TARGET)

test: $(TARGET)
        ./tests/test.sh

clean:
        rm -f $(TARGET)

.PHONY: all run test clean
```

GNU nano 8.7.1 tests/test.sh *

```
#!/bin/bash

PROGRAM="./skill24"

echo "Running Skill 24 tests..."
echo "-------------------------"

if [ ! -f "$PROGRAM" ]; then
    echo "FAIL: Executable not found."
    exit 1
fi

OUTPUT=$($PROGRAM)

if echo "$OUTPUT" | grep -q "Repository validation successful."; then
    echo "PASS: Repository validation test"
else
    echo "FAIL: Repository validation test"
    exit 1
fi

if echo "$OUTPUT" | grep -q "Version: 1.0.0"; then
    echo "PASS: Version test"
else
    echo "FAIL: Version test"
    exit 1
fi

echo "-------------------------"
echo "All tests passed."
```

GNU nano 8.7.1 README.md *

````
# Skill 24 - Repository Organization

## Project Description

This project demonstrates repository organization,
build automation, testing, documentation, and
release tagging.

## Project Structure

```text
Skill24_Project/
├── Makefile
├── README.md
├── docs/
│   └── architecture.txt
├── include/
│   └── config.h
├── src/
│   └── main.c
└── tests/
    └── test.sh
````
````

```
  GNU nano 8.7.1                                          docs/architecture.txt *
SKILL 24 ARCHITECTURE

Source Code
    |
    v
src/main.c
    |
    v
include/config.h
    |
    v
Makefile
    |
    v
Executable
    |
    v
Automated Tests
    |
    v
Git Repository
    |
    v
Release Tag v1.0.0
```

```
sanjana3027@DESKTOP-QJF5AGU:        ; git remote add origin <https://sanjana3027@DESKTOP-QJF5AGU:                PRACTICAL_OS_2520030487/tree/main/SKILL>
-bash: syntax error near unexpected token `newline'
sanjana3027@DESKTOP-QJF5AGU:        ; git remote add origin https://github.com/ketanasindhusomisetty/SKILL_PRACTICAL_OS_2520030487/tree/main/SKILL
sanjana3027@DESKTOP-QJF5AGU:        ; git remote -v
origin  https://github.com/ketanasindhusomisetty/SKILL_PRACTICAL_OS_2520030487/tree/main/SKILL (fetch)
origin  https://github.com/ketanasindhusomisetty/SKILL_PRACTICAL_OS_2520030487/tree/main/SKILL (push)
sanjana3027@DESKTOP-QJF5AGU:        ;
```

```
 * [new tag]          v1.0.0 -> v1.0.0
sanjana3027@DESKTOP-QJF5AGU:       /Skill_Project$ git status
On branch skill24
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
sanjana3027@DESKTOP-QJF5AGU:       /Skill_Project$
```